

against vulnerabilities:

- Tests for exploitation and manipulation in the communication ports
- IoT sniffing by capture and analysis of radio signals
- Detection of interfaces or backdoors
- Tests for buffer overflow
- Tests for breaking of passwords
- Debugging
- Cryptographic analysis
- Manipulation in firmware tests
- Reverse engineering
- Memory dumps

Open source tools for the security audit of IoT devices

IoT devices are becoming more and more commonplace in our daily lives. A few examples of such devices are smart bicycles, fitness trackers, medical sensors, smart locks, and linked factories. All of these devices and gadgets can be secured against cyber attacks using open source tools, some of which are briefly discussed here.

PENIOT

<https://github.com/yakuza8/peniot>
PENIOT is an IoT penetration testing tool that enables security audit teams to test and breach the devices with various security threats by exploiting their connectivity. There are both active and passive security threats that can be tested. It is possible to carry out active security attacks, such as altering system resources, replaying legitimate communication units and so on, after determining the target device and the relevant information (or parameters). Passive security threats, such as breaking the confidentiality of sensitive data or gaining access to network traffic analysis, can also be analysed.

Objection

<https://github.com/sensepost/objection>
Objection is a tool for the detailed analysis and security audit of the

Android and iOS apps used in the IoT environment.

Nowadays, a number of smart gadgets and devices are using Android and iOS platforms which can be analysed with detailed logs and security audit reports, using this tool.

Routersploit

<https://github.com/threat9/routersploit>

This powerful open source exploitation framework for embedded devices has multiple features and modules for penetration testing and security audit:

- Exploits - vulnerabilities evaluation
- Creds - testing of network services and credentials
- Scanners - detailed security audit of target
- Payloads - generation of payloads and injection key points
- Generic - performing and testing of assaults

Wireshark

<https://www.wireshark.org>

Wireshark is a free network protocol analyser with a lot of features. Various protocols, including MQTT, are used by IoT devices to communicate and these can be analysed effectively. Security rules are configured according to the protocol and the traffic is examined in order to discover any weaknesses. A network packet analyser

is accessible through the command line using tcpdump. Such tools are used to examine data packets exchanged between IoT devices and networks.

Binwalk

<https://www.kali.org/tools/binwalk>

Binwalk is a tool for reversing the design of hardware. It is one of the key components in Kali Linux that is used for penetration testing, server fingerprinting, security audit and forensic applications.

Firmwalker

<https://github.com/craigz28/firmwalker>

Firmwalker is a free and open source tool for searching and scanning the firmware file system, irrespective of whether it is extracted or mounted. A detailed security audit can be done using this tool.

In the era of Internet of Things (IoT) and Internet of Everything (IoE), there is a need to devise and use high performance toolkits for penetration testing and security audits. As the number of IoT devices grows, so do the security risks. Free and open source toolkits and software suites should be customised according to recent protocols and dynamic traffic so that higher levels of security and privacy can be enforced in IoT and IoE deployments. 

References

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